Prepare DP1 paper (SP-2021)

[SP-2834] Respond to review feedback on Section 5: Performance

Created: 05/Dec/25 Updated: 12/Feb/26 Resolved: 12/Feb/26

Status:	Done
Project:	System Performance
Components:	None
Fix versions:	None
Parent:	Prepare DP1 paper

Type:	Story	**Priority:**	Undefined
Reporter:	Leanne Guy	**Assignee:**	Unassigned
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Reviewers:	James Mullaney
Sprint:	

Description

Section 5.11 - The text describes significant issues in running the deblending pipeline because of the stellar crowding due to memory or runtime limits being exceeded in the 47 Tucanae and Fornax. I appreciate that at some level of extreme stellar backgrounds the pipelines will fail, but there is little discussion of what conditions the pipelines are supposed to successfully run in terms of crowded field photometry for DP1. Were the pipelines expected to be able to successfully run deblending on these fields? Is there an expected/estimated galactic latitude limit/threshold where performance of the pipelines on single visits/coaads would expect to successfully run for the version of the pipelines that processed DP1?

Section 5.4 - Please comment on or show the distribution of airmasses for the 228 visits used in the DCR analysis.

Figure 24 - "The g− i color is computed for every source in the reference catalog that is matched to a direct source in the science image, and the binned density for the full survey..." What reference catalog is used for this analysis? Is this reference catalog the DP1 object catalog or is some other catalog

that's been photometrically transformed to g-i in the LSSTCam filter bands? Further clarification in the text about how this figure was generated is needed.

Section 5.6 - I think a figure showing the completenesses from the Hubble Legacy Field and Euclid Q1 would be helpful with the caveats referenced in the main text added to the figure caption.

Section 5.8 - How was the labeling carried out. Was only person assigned to review each of the 9500 DIASource image triplets? How many people were involved in reviewing the images. How big was the cutout that the review looked at. It would be helpful to see what the team was reviewing as a figure.

Comments

Comment by Leanne Guy [28/Jan/26]

Section 5.11 - The text describes significant issues in running the deblending pipeline because of the stellar crowding due to memory or runtime limits being exceeded in the 47 Tucanae and Fornax. I appreciate that at some level of extreme stellar backgrounds the pipelines will fail, but there is little discussion of what conditions the pipelines are supposed to successfully run in terms of crowded field photometry for DP1. Were the pipelines expected to be able to successfully run deblending on these fields? Is there an expected/estimated galactic latitude limit/threshold where performance of the pipelines on single visits/coaads would expect to successfully run for the version of the pipelines that processed DP1?

> DP1 is a preview and was the first testing ground for the pipelines, and have given us an understanding of where they breakdown. This is important information to improve the processing.
>
> This is stated in the text.
> Although the DP1 pipeline was not optimized for crowded-field photometry, these early studies … highlight both the limitations and opportunities for science with Rubin data in crowded environments, and they inform future pipeline development aimed at robust source recovery in complex stellar fields
>
> I will not make any changes

Section 5.4 - Please comment on or show the distribution of airmasses for the 228 visits used in the DCR analysis.

> Added statement for range and mea of airmasses

Figure 24 - "The g− i color is computed for every source in the reference catalog that is matched to a direct source in the science image, and the binned density for the full survey..." What reference catalog is used for this analysis? Is this reference catalog the DP1 object catalog or is some other catalog that's been photometrically transformed to g-i in the LSSTCam filter bands? Further clarification in the text about how this figure was generated is needed.

> The monster – clarfied in paper

Section 5.6 - I think a figure showing the completenesses from the Hubble Legacy Field and Euclid Q1 would be helpful with the caveats referenced in the main text added to the figure caption.

> It would, but that is beyond the scope of this paper and is left to the science community.

Section 5.8 - How was the labeling carried out. Was only person assigned to review each of the 9500 DIASource image triplets? How many people were involved in reviewing the images. How big was the cutout that the review looked at. It would be helpful to see what the team was reviewing as a figure.

> Description added (but committed on SP-2833 DONE) of the labelling

Comment by Leanne Guy [29/Jan/26]

This branch should also respond to
Section 4.6.3 - I would expect some sort of orbit table tallying the number of objects in different types of orbits or discussion of what was found for moving Solar System. Were all the discoveries Main Belt Asteroids? As there is a Minor Planet Center orbit table/catalog this information must be available. Please add some summary statistics and figures about the Solar System objects associated and new objects found in DP1.

> 4.6.3 is the DRP section and describes the processing. This is a result and would go in section 5.9 : Solar System ….. Response will be addeed on https://rubinobs.atlassian.net/browse/SP-2834

They're all main belt asteroids. There are few enough of them and orbits are poorly-enough constrained by DP1 that a plot of their orbits wouldn't be more informative than the phrase "93 new main belt asteroids".

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